MLA-C01

AWS CERTIFIED

COURSE SLIDES · WSQ

AWS Certified Machine Learning Engineer Associate Training

WSQ Course Code: TGS-2024049340

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v14 · 23 August 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

Mandatory each session

Take AM, PM and Assessment digital attendance for WSQ-funded courses.

2

TRAQOM QR code

The trainer displays the SSG digital-attendance QR code at the start of each session.

3

Scan to submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

At least 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

CERTIFICATION
AWS-certified; cloud, machine learning and data-engineering practitioner and trainer.

DELIVERS
WSQ courses on AWS, machine learning, data engineering, AI and cloud architecture.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with AWS, Python, data or machine learning (if any).
- What you want to be able to build or certify after this course.

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HOUSEKEEPING

Ground Rules

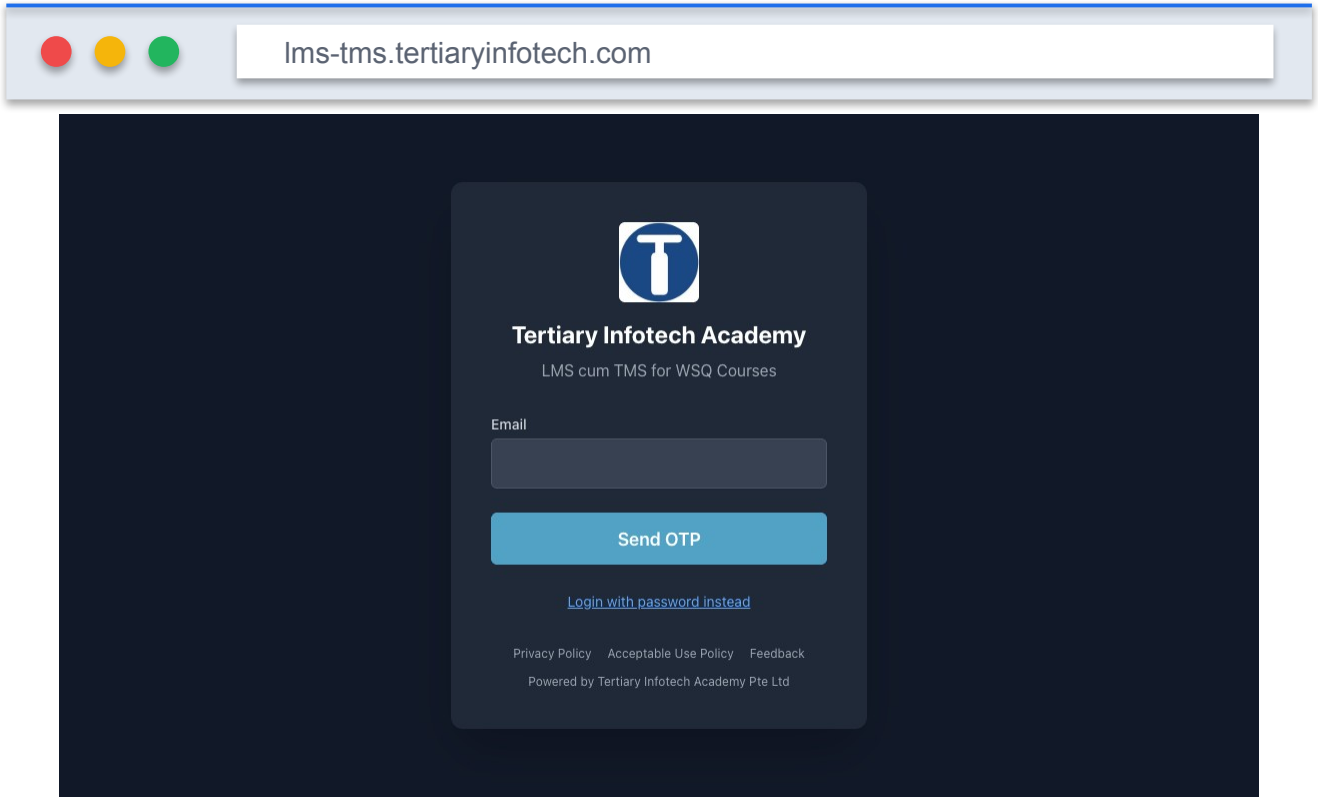
1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL · LMS-TMS

Download Course Material

1. Open lms-tms.tertiaryinfotech.com in your browser.
2. Log in with the account emailed to you before the course.
3. Open this course and go to the Courseware / Materials tab.
4. Download the Trainer Slides, Learner Guide and Lesson Plan.
5. Keep them open — the assessment is open book.



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SCHEDULE

Lesson Plan — 3 Days · 24 Course Hours

1

Day 1 — Data Preparation for Machine Learning

Domain 1 · Labs 1–4: account safety & IAM, S3 data lake, batch/streaming ingestion, feature engineering.

2

Day 2 — ML Model Development

Domain 2 · Labs 5–8: modelling approach, training & Debugger, tuning & registry, evaluation & bias.

3

Day 3 — Deployment, Monitoring, Security & Exam Review

Domains 3–4 · Labs 9–12: deployment, MLOps, monitoring, security, cost & exam review.

4

Assessment (Day 3)

Written Assessment (1 hour) + Case Study (1 hour) = 2 assessment hours.

5

Duration

22 learning hours + 2 assessment hours = 24 course hours across 3 days.

6

Daily timing

9:30am–6:30pm · 1-hour lunch · two 15-min tea breaks within the 8 scheduled hours.

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Data Preparation

Ingest, transform, engineer features for, and govern ML data on AWS.

2

LO2 · Model Development

Select, train, tune, evaluate and register models with SageMaker AI.

3

LO3 · Deploy & Orchestrate

Real-time, serverless and batch inference; MLOps pipelines and CI/CD.

4

LO4 · Monitor & Secure

Monitoring, security, cost optimisation and MLA-C01 exam readiness.

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WSQ FUNDING

Criteria for Funding

1

Attendance

Attend at least 75% of the course.

2

Competency

Be assessed Competent (C) in both the WA and the Case Study (CS).

3

Digital attendance

Take AM, PM and Assessment digital attendance (TRAQOM) each day.

4

Survey

Complete the TRAQOM course survey.

5

Eligibility

Meet SSG/SkillsFuture eligibility for the applicable funding scheme.

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BEFORE THE ASSESSMENT

Briefing for Assessment

1. Place phones and other materials under the table or on the floor.
2. No photos or recording of assessment scripts.
3. No discussion during the assessment.
4. Use a black/blue pen for hard-copy assessments.
5. No liquid paper / correction tape.
6. Scripts are collected when time is up.

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FINAL ASSESSMENT

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book. Knowledge questions trace to specific course slides.
- Case Study (CS) — an applied AWS ML scenario, 1 hour, open book. Learners analyse a business scenario and design and justify an end-to-end ML solution mapped to the four MLA-C01 domains and the twelve hands-on labs.
- Format: Open Book — slides, Learner Guide and approved materials only.
- Held on Day 3: Written Assessment (1 hour) then Case Study (1 hour) = 2 assessment hours.
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- An appeal process is available if required.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then Case Study (CS) — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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CORE CONCEPTS

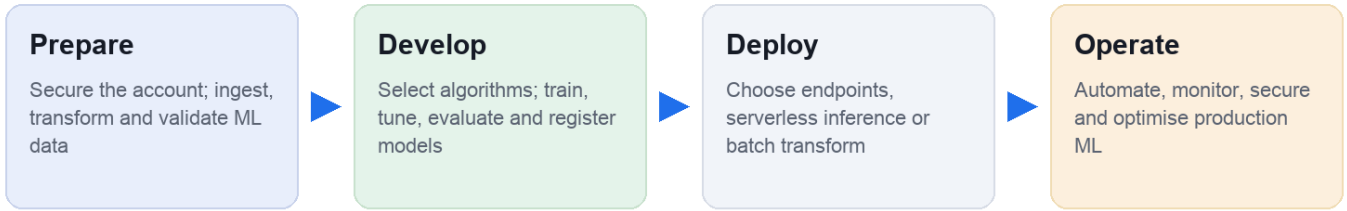
AWS Machine Learning Engineering

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THE COURSE AT A GLANCE

AWS Machine Learning Engineering Lab Flow

AWS Machine Learning Engineering Lab Flow



All diagrams are aligned to the twelve AWS hands-on labs and reused across the PPT, Learner Guide and Lesson Plan.

Prepare → Develop → Deploy → Operate — the twelve labs follow the ML lifecycle.

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END TO END

The Machine Learning Lifecycle on AWS

1

1 · Prepare data

Ingest, store, transform and validate data; engineer and store features (Domain 1).

2

2 · Develop model

Choose an approach; train, tune, evaluate, and register the model (Domain 2).

3

3 · Deploy & orchestrate

Serve inference and automate the pipeline with MLOps (Domain 3).

4

4 · Monitor & secure

Monitor drift, secure, and cost-optimise in production (Domain 4).

5

Repeatable

SageMaker Pipelines, Model Registry and IaC make each stage repeatable.

6

Exam-aligned

Every stage maps to an MLA-C01 domain and a hands-on lab.

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THE SERVICE STACK

AWS Services You Will Practise

AWS Services Practised



All diagrams are aligned to the twelve AWS hands-on labs and reused across the PPT, Learner Guide and Lesson Plan.

Data, Model, Deploy and MLOps services used across the twelve labs.

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Data Types & Formats for ML

1

Structured

Tabular rows/columns — CSV, Parquet, ORC for analytics and SageMaker training.

2

Semi-structured

JSON, Avro — flexible schemas, common for ingestion and streaming.

3

Unstructured

Text, image, audio, video — for NLP, computer vision and generative AI.

4

Format choice

Columnar (Parquet/ORC) for analytics; RecordIO/protobuf for efficient training.

5

Storage

Amazon S3 data lake with lifecycle prefixes and default encryption.

6

Governance

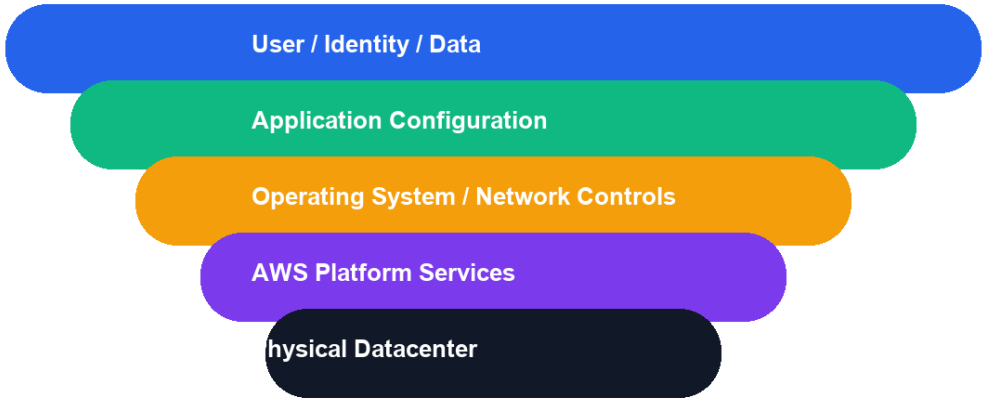
Glue Data Catalog, Lake Formation and Clarify for quality and bias.

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SECURITY MODEL

Shared Responsibility for ML on AWS

AWS Shared Responsibility Layers



AWS MLA-C01 labs show which controls learners configure and which controls AWS operates.

AWS secures the cloud; you secure your ML data, models and access in the cloud.

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WHAT THE EXAM COVERS

MLA-C01 Exam Domains & Weightings

AWS MLA-C01 Domain Weight Matrix

Domain	Exam Weight	Labs	Practice Priority
Data Preparation	28%	1–4	High
Model Development	26%	5–8	High
Deployment & Orchestration	22%	9–10	High
Monitoring & Security	24%	11–12	High

Use this matrix to prioritise revision and lab recap time before assessment.

Four domains — the course spends time in proportion to their exam weighting.

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01

DOMAIN 01

Data Preparation for Machine Learning

Ingestion & storage · transformation & feature engineering · data quality, labelling & bias

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MLA-C01 EXAM WEIGHTING 28%

Key Concepts — Data Preparation for Machine Learning

1

Data formats

CSV, JSON, Parquet, ORC, Avro and RecordIO — column vs row storage, and which format suits analytics, streaming and SageMaker training.

2

S3 ML data lake

Bucket + prefix design (raw/prepared/features/training/models/monitoring), default encryption (SSE-S3 / SSE-KMS) and Block Public Access.

3

Batch & streaming ingestion

AWS Glue crawlers/jobs and the Data Catalog, Athena, and the Kinesis family (Data Streams, Firehose, MSK) for batch and streaming preparation.

4

Feature engineering

Imputation, normalisation/standardisation, binning, one-hot encoding and scaling; online vs offline SageMaker Feature Store.

5

Labelling

SageMaker Ground Truth labelling jobs for text, image and tabular data; human-in-the-loop and automated labelling.

6

Data quality & bias

Glue Data Quality and SageMaker Clarify checks for nulls, duplicates, schema drift, outliers, class imbalance and data leakage.

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ML Data Formats Compared

Format	Layout	Best used for
CSV	Row, text	Human-readable ingestion and small tabular datasets
JSON	Semi-structured	APIs, nested records and event/log data
Parquet	Columnar, compressed	Analytics and large-scale SageMaker training
ORC	Columnar, compressed	Hive/analytics workloads with predicate push-down
Avro	Row + embedded schema	Streaming and schema evolution
RecordIO-protobuf	Binary	Efficient input to several SageMaker built-in algorithms

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Data Types for Machine Learning

1

Structured

Tabular rows/columns — CSV, Parquet, ORC for analytics and training.

2

Semi-structured

JSON and Avro — flexible, nested schemas common in ingestion.

3

Unstructured

Text, image, audio, video for NLP, vision and generative AI.

4

Labelled vs unlabelled

Supervised needs labels (Ground Truth); unsupervised does not.

5

Time-series

Ordered observations for forecasting and anomaly detection.

6

Streaming vs at-rest

Live events vs data already landed in the S3 data lake.

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Amazon S3 Encryption Options

Option	Key management	Choose when
SSE-S3	S3-managed AES-256	A simple, always-on default for training data
SSE-KMS	AWS KMS (CMK)	You need audit logs, rotation and access control on the key
SSE-C	Customer-supplied	You must hold and provide the key on each request
Client-side	Encrypted before upload	Data must be encrypted before it reaches AWS
TLS in transit	HTTPS/SSL	Always — protects data moving to and from S3

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S3 Storage Classes & Lifecycle for ML

1

S3 Standard

Hot raw and training data accessed frequently.

2

Intelligent-Tiering

Unknown/changing access — S3 moves objects automatically.

3

Standard-IA / One Zone-IA

Prepared datasets kept but read infrequently.

4

Glacier Instant / Flexible

Archived training snapshots and old model artifacts.

5

Lifecycle rules

Transition raw→IA→Glacier and expire temp objects on a schedule.

6

Block Public Access

Keep enabled on every ML bucket to prevent data exposure.

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IAM & Access Control for ML

1

Users, groups, roles

Prefer roles over long-lived keys; group common permissions.

2

Least privilege

Grant only the S3/SageMaker/Glue actions each task needs.

3

SageMaker execution role

The role SageMaker assumes to read S3 and write artifacts.

4

MFA

Protect human sign-in with multi-factor authentication.

5

Policies & conditions

Identity, resource and permission-boundary policies with conditions.

6

KMS key policies / grants

Control who can use the CMK that encrypts ML data.

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Batch vs Streaming Ingestion

Aspect	Batch	Streaming
Services	Glue jobs, EMR (Spark), Athena, DMS	Kinesis Data Streams, Data Firehose, Managed Kafka
Latency	Minutes to hours	Seconds — near real time
Typical use	Nightly ETL, historical reprocessing	Clickstream, IoT, live features
Lands in	S3 data lake, cataloged by Glue	S3/Firehose → Glue crawler → Catalog

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AWS Glue Components

1

Crawlers

Scan S3, infer schema and populate the Data Catalog.

2

Data Catalog

Central table/partition metadata shared by Athena, EMR, SageMaker.

3

Glue ETL jobs

Serverless Spark jobs that clean and reshape data to Parquet.

4

Glue DataBrew

Visual, no-code profiling and cleaning of datasets.

5

Glue Data Quality

Rule-based checks for nulls, ranges, uniqueness and freshness.

6

Triggers & workflows

Schedule or event-chain crawlers and jobs into a pipeline.

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Streaming Ingestion Family

Service	Does	Best for
Kinesis Data Streams	Durable, shard-based real-time stream	Custom consumers, sub-second processing
Kinesis Data Firehose	Managed delivery to S3/Redshift/OpenSearch	No-code load of streams into the lake
Managed Service for Kafka	Managed Apache Kafka (MSK)	Kafka-based ecosystems and high throughput
Managed Flink	Real-time analytics on streams	In-stream aggregation and feature computation

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Feature Engineering Transforms

1. **Missing-value imputation**
   Fill nulls with mean/median/mode or a model.
2. **Scaling / normalisation**
   Standardise or min-max scale numeric features.
3. **Encoding categoricals**
   One-hot, label or target encoding for categories.
4. **Binning & bucketing**
   Group continuous values into ranges.
5. **Outlier handling**
   Cap, clip or remove extreme values.
6. **Avoid data leakage**
   Never use future/target-derived information as a feature.

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SageMaker Feature Store — Online vs Offline

Store	Latency	Used for
Online store	Single-digit ms	Real-time inference feature lookups
Offline store	Batch (S3)	Training datasets and historical analysis
Feature groups	—	Versioned schema of related features
Consistency	—	Same feature definition for train and serve

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DOMAIN 01 · TRUSTWORTHY DATA

Labelling, Data Quality & Bias

1. **Ground Truth**
   Managed labelling for text, image and tabular data.

2. **Human-in-the-loop**
   A2I adds human review to low-confidence predictions.

3. **Automated labelling**
   Active learning labels the easy cases to cut cost.

4. **Glue Data Quality**
   Automated rule checks on nulls, ranges and freshness.

5. **Clarify pre-training bias**
   Detect class imbalance and skew before training.

6. **Data-leakage checks**
   Remove target/future signals that inflate scores.

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Choosing a Data-Preparation Service

Visual, no-code cleaning and profiling	**AWS Glue DataBrew**
Interactive feature engineering for ML	**SageMaker Data Wrangler**
Large-scale distributed ETL (Spark)	**AWS Glue jobs or Amazon EMR**
Ad-hoc SQL exploration of data in S3	**Amazon Athena**
Near-real-time event ingestion into S3	**Kinesis Data Firehose**
Reusable features shared train/serve	**SageMaker Feature Store**
Data labelling for text/image/tabular	**SageMaker Ground Truth**

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Amazon Athena & the Glue Data Catalog

1

Serverless SQL

Query S3 data directly with standard SQL — no servers.

2

Glue Data Catalog

Shared table/partition metadata for Athena and SageMaker.

3

Partitioning

Partition by date/key to cut scan cost and time.

4

Columnar wins

Parquet/ORC scan far less data than CSV.

5

CTAS / views

Create prepared tables and reusable views for ML.

6

Cost model

You pay per terabyte scanned — prune columns and partitions.

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SageMaker Data Wrangler Transforms

1. **Import**
   Connect to S3, Athena, Redshift and more.
2. **Data insights**
   Automatic profiling and quality warnings.
3. **300+ transforms**
   Built-in cleaning, encoding and scaling steps.
4. **Custom transforms**
   Add pandas/PySpark/SQL steps where needed.
5. **Export**
   Push the flow to a processing job or pipeline.
6. **Feature Store target**
   Write engineered features straight to Feature Store.

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WHAT YOU'LL DO

Hands-On Labs — Data Preparation for Machine Learning

Labs 1–2

- Account Safety and IAM Baseline
- S3 ML Data Lake and Data Formats

Labs 3–4

- Batch and Streaming Ingestion with Glue, Athena and Kinesis
- Feature Engineering, Feature Store, Labelling and Data Quality

—

- —

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DOMAIN 01 · HANDS-ON BRIEFING

Account Safety and IAM Baseline

LAB 1

Set up a safe AWS baseline for ML engineering: confirm identity and Region, review IAM users, roles, groups and least privilege, understand the SageMaker execution role, enable MFA and a budget alert, and review KMS key concepts before any data lands.

You'll build

A secured AWS ML workspace: verified identity/Region, a least-privilege IAM review, an enabled budget alert and notes on execution roles and KMS.

Tools: IAM, AWS KMS, AWS Budgets, Amazon SageMaker AI, AWS CloudShell, AWS CloudTrail

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Workflow — Account Safety and IAM Baseline

1. Sign in & verify identity/Region
2. Review IAM roles & least privilege
3. Review the SageMaker execution role
4. Enable MFA & a budget alert
5. Review KMS key concepts

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LAB 1 · KEY DECISIONS

Trade-offs — Account Safety and IAM Baseline

Use IAM roles when

- Granting AWS services access (execution roles)
- Temporary, auto-rotated credentials
- Cross-account or federated access

Use IAM users/keys when

- A human needs console sign-in
- No role-assumption path is available
- Always protect with MFA and rotation

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LAB 1 · VERIFY & TROUBLESHOOT

Account Safety and IAM Baseline

✅ Test it — acceptance criteria

You can verify your identity and Region, explain least-privilege IAM and the SageMaker execution role, and confirm MFA, a budget alert and KMS are in place before any ML data lands.

Troubleshooting

Access denied calling an ML API	Attach a least-privilege policy for that action to your role/user.
SageMaker cannot read S3	Fix the SageMaker execution role's S3 permissions and trust policy.
Surprise charges appear	Enable AWS Budgets alerts and delete idle resources immediately.

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DOMAIN 01 · HANDS-ON BRIEFING

S3 ML Data Lake and Data Formats

LAB 2

Build the storage foundation for ML: create an encrypted S3 data lake with raw, prepared, features, training, models and monitoring prefixes, apply Block Public Access, lifecycle rules and storage classes, and compare the data formats used across ML workflows.

You'll build

A secure, encrypted S3 ML data lake with six lifecycle prefixes and a data-format comparison table.

Tools: Amazon S3, AWS KMS, S3 Lifecycle, Amazon SageMaker AI

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LAB 2 · ARCHITECTURE & WORKFLOW

Workflow — S3 ML Data Lake and Data Formats

1. Create an encrypted S3 bucket
2. Enable Block Public Access
3. Add ML lifecycle prefixes
4. Set storage-class lifecycle rules
5. Compare data formats

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LAB 2 · KEY DECISIONS

Trade-offs — S3 ML Data Lake and Data Formats

Use SSE-KMS when

- You need key rotation and audit logs
- Fine-grained key access control
- Regulated data (PII/PHI)

SSE-S3 is enough when

- A simple always-on default is fine
- No per-key access control needed
- Lowest operational overhead

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LAB 2 · VERIFY & TROUBLESHOOT

S3 ML Data Lake and Data Formats

✅ Test it — acceptance criteria

You can navigate the encrypted S3 data lake with its six prefixes, explain storage-class lifecycle, and justify a data-format choice for analytics, streaming and SageMaker training.

Troubleshooting

Bucket name already exists	S3 names are globally unique — add your initials/date.
Access denied on PutObject	Attach a least-privilege S3 policy to your role/execution role.
Objects not encrypted	Set default bucket encryption; existing objects must be re-copied.

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DOMAIN 01 · HANDS-ON BRIEFING

Batch and Streaming Ingestion with Glue, Athena and…

LAB 3

Land data into the lake two ways: crawl and catalog batch data with AWS Glue and query it with Athena, and design a near-real-time streaming path with the Kinesis family (Data Streams, Firehose, MSK) into S3, then compare batch and streaming trade-offs.

You'll build

A cataloged batch dataset queryable in Athena plus a streaming-ingestion design and a batch-vs-streaming comparison.

Tools: AWS Glue, Glue Data Catalog, Amazon Athena, Kinesis Data Streams, Kinesis Data Firehose, Amazon MSK, Amazon S3

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LAB 3 · ARCHITECTURE & WORKFLOW

Workflow — Batch and Streaming Ingestion with…

1. Crawl raw/ with Glue
2. Catalog the schema
3. Query in Athena
4. Design a Kinesis streaming path
5. Compare batch vs streaming

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LAB 3 · KEY DECISIONS

Trade-offs — Batch and Streaming Ingestion with…

Batch ingestion (Glue/EMR)

- Scheduled, large-scale Spark ETL
- Writes columnar Parquet to S3
- Best for full historical loads

Streaming ingestion (Kinesis)

- Near-real-time event capture
- Firehose delivers to S3 with no code
- Best for clickstream/IoT/live features

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LAB 3 · VERIFY & TROUBLESHOOT

Batch and Streaming Ingestion with Glue, Athena and…

✅ Test it — acceptance criteria

You can catalog and query batch data with Glue and Athena, design a Kinesis streaming path into the lake, and justify batch vs streaming from latency and use-case requirements.

Troubleshooting

Crawler finds no tables	Point the crawler at the correct S3 prefix and re-run it.
Athena query fails	Set an S3 query-result location and confirm the Glue table exists.
Streaming data not landing in S3	Check the Firehose delivery stream's IAM role and S3 destination.

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DOMAIN 01 · HANDS-ON BRIEFING

Feature Engineering, Feature Store, Labelling and…

LAB 4

Turn cataloged data into trustworthy features: apply feature-engineering transforms in Data Wrangler, store reusable features in SageMaker Feature Store, review Ground Truth labelling, and validate quality and bias with Glue Data Quality and SageMaker Clarify.

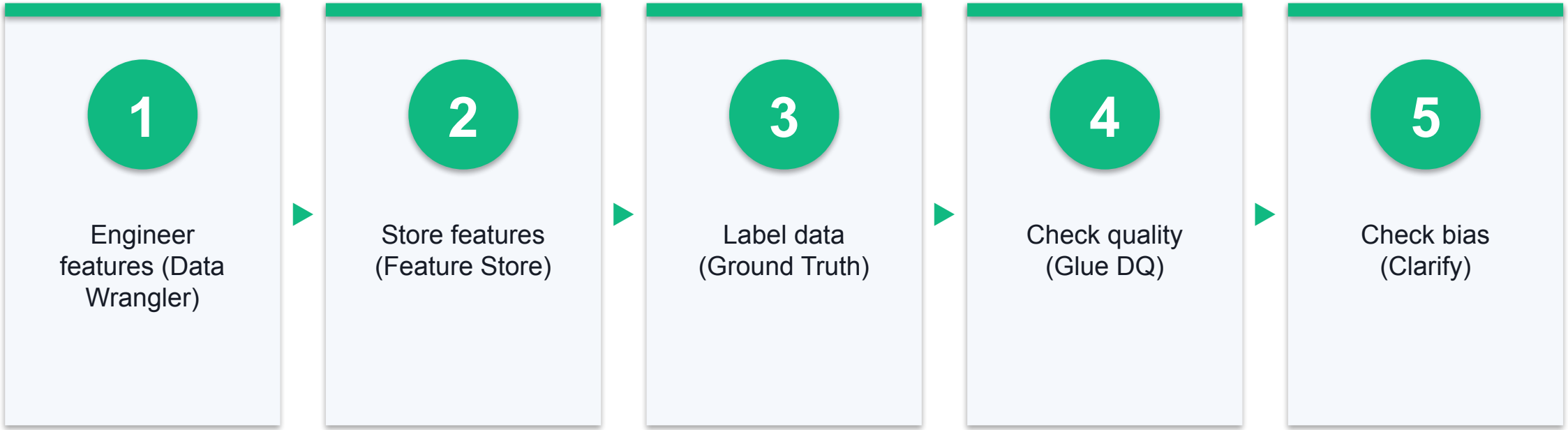
You'll build

A feature-engineering flow, an online/offline Feature Store design, and feature-engineering, data-quality and bias checklists.

Tools: SageMaker Data Wrangler, Feature Store, Ground Truth, Clarify, AWS Glue, Glue Data Quality, Amazon Athena

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Workflow — Feature Engineering, Feature Store,…



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LAB 4 · KEY DECISIONS

Trade-offs — Feature Engineering, Feature Store,…

Interactive prep (Data Wrangler)

- Visual feature engineering on samples
- Fast iteration and preview
- Export the flow into a pipeline

Reusable features (Feature Store)

- Online store for low-latency serving
- Offline store for training
- Same definition train and serve

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LAB 4 · VERIFY & TROUBLESHOOT

Feature Engineering, Feature Store, Labelling and…

✅ Test it — acceptance criteria

You can engineer features, store them for reuse in Feature Store, describe Ground Truth labelling, and validate a dataset for quality and bias with Glue Data Quality and Clarify.

Troubleshooting

Features differ between train and serve	Use Feature Store so both read the same definition.
Skewed classes bias the model	Detect with Clarify; rebalance or reweight before training.
Data-quality rules keep failing	Fix the source or the transform; re-run Glue Data Quality.

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DOMAIN RECAP

Recap — Data Preparation for Machine Learning

- You can now: secure an AWS ML workspace with identity, least-privilege IAM and cost guardrails (Domain 1: establish a safe ML foundation)
- You can now: structure and secure an S3 data lake for ML and choose the right data format (Domain 1: data ingestion and storage)
- You can now: design batch and streaming ingestion into the ML data lake and query it in place (Domain 1: ingest and catalog data)
- You can now: engineer, store, label and validate features for machine learning (Domain 1: transform, engineer and validate data)

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02

DOMAIN 02

ML Model Development

Modelling approach · training & debugging · tuning, experiments & Model Registry · evaluation, metrics & bias

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MLA-C01 EXAM WEIGHTING 26%

Key Concepts — ML Model Development

1

Problem framing

Classification, regression, clustering, forecasting, anomaly detection, NLP and computer vision — matching the problem to an approach.

2

Approach selection

SageMaker built-in algorithms vs script-mode custom frameworks vs JumpStart vs Amazon Bedrock foundation models vs managed AI services.

3

Training & debugging

Training jobs: input channels, instance type, container image, execution role; SageMaker Debugger and Profiler to catch training problems early.

4

Tuning & experiments

Automatic Model Tuning (random, grid, Bayesian, Hyperband); SageMaker Experiments for reproducible trial tracking.

5

Model Registry

Model package groups, versions, approval status and metadata for a repeatable train-to-approval workflow.

6

Evaluation & bias

Confusion matrix, accuracy, precision, recall, F1, ROC AUC; MAE/MSE/RMSE/R2; Clarify bias metrics, SHAP and feature importance.

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ML Problem Types

1. **Classification**
   Predict a category — churn, spam, fraud (supervised).
2. **Regression**
   Predict a number — price, demand, load (supervised).
3. **Clustering**
   Group similar records — segmentation (unsupervised).
4. **Forecasting**
   Predict future values of a time-series (DeepAR).
5. **Anomaly detection**
   Flag rare/outlier events — Random Cut Forest.
6. **NLP & computer vision**
   Text and image understanding and generation.

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SageMaker Built-in Algorithms

Algorithm	Problem type	Typical use
XGBoost	Classification / regression	Tabular churn, risk, price prediction
Linear Learner	Classification / regression	Fast linear baselines at scale
K-Means	Clustering	Customer/segment grouping (unsupervised)
Random Cut Forest	Anomaly detection	Fraud and outlier detection
DeepAR	Time-series forecasting	Demand and load forecasting
BlazingText / Seq2Seq	NLP	Text classification and translation
Image Classification / Object Detection	Computer vision	Labelling and detecting objects in images

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Choosing a Modelling Approach

Common tabular/CV/NLP task, want managed algorithm	▶	**SageMaker built-in algorithm**
Custom TensorFlow / PyTorch / scikit-learn code	▶	**SageMaker script mode**
Full control of runtime and dependencies	▶	**Bring Your Own Container (ECR)**
Start from a pretrained/open model, fine-tune	▶	**SageMaker JumpStart**
Generative text/summarisation via foundation model	▶	**Amazon Bedrock**
Standard perception task, no training wanted	▶	**Managed AI service (Rekognition, Comprehend, Textract…)**

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Managed AI Services

1. **Amazon Rekognition**
   Image and video analysis — objects, faces, moderation.

2. **Amazon Comprehend**
   NLP — entities, sentiment, key phrases, PII detection.

3. **Amazon Textract**
   Extract text, forms and tables from documents.

4. **Amazon Translate**
   Neural machine translation between languages.

5. **Amazon Transcribe / Polly**
   Speech-to-text and text-to-speech.

6. **Amazon Bedrock**
   Foundation models for text, chat, summarisation and RAG.

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SageMaker Training Job Anatomy

1

Input channels

train/validation/test data read from S3 (or Feature Store).

2

Instance & count

Instance type/family and how many for the job.

3

Container image

Built-in, framework (script mode) or your own (BYOC).

4

Execution role

IAM role SageMaker assumes to access S3 and write output.

5

Hyperparameters

Epochs, batch size, learning rate passed to the job.

6

Output path

Model artifact (model.tar.gz) written back to S3.

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Training Concepts & Input Modes

1

Epoch / batch / learning rate

One full pass; samples per step; step size toward the minimum.

2

Early stopping

Stop when validation stops improving to avoid overfitting and cost.

3

File mode

Copies the full dataset to the instance before training starts.

4

Pipe / FastFile mode

Streams data from S3 — faster start, less local disk.

5

Distributed training

Data/model parallel across instances for large jobs.

6

Managed Spot training

Up to ~90% cheaper with checkpointing for interruptions.

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DOMAIN 02 · CATCH TRAINING PROBLEMS

SageMaker Debugger & Profiler

1

Tensor capture

Save weights, gradients and losses during training.

2

Built-in rules

Detect vanishing/exploding gradients, overfitting, stalls.

3

Profiler

Spot GPU/CPU under-utilisation and I/O bottlenecks.

4

Actions & alerts

Stop a broken job automatically to save cost.

5

TensorBoard

Visualise curves to diagnose training behaviour.

6

Why it matters

Fail fast instead of paying for a doomed long job.

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Automated & Assisted Model Development

1

SageMaker Autopilot

AutoML — explores algorithms/features and ranks candidate models.

2

Automatic Model Tuning

Random, grid, Bayesian or Hyperband search over an objective metric.

3

SageMaker Experiments

Tracks trials, parameters and metrics for reproducibility.

4

SageMaker Debugger

Captures tensors to catch vanishing gradients, overfitting, stalls.

5

SageMaker Canvas

No-code model building for business analysts.

6

SageMaker JumpStart

Prebuilt models and solution templates to deploy or fine-tune.

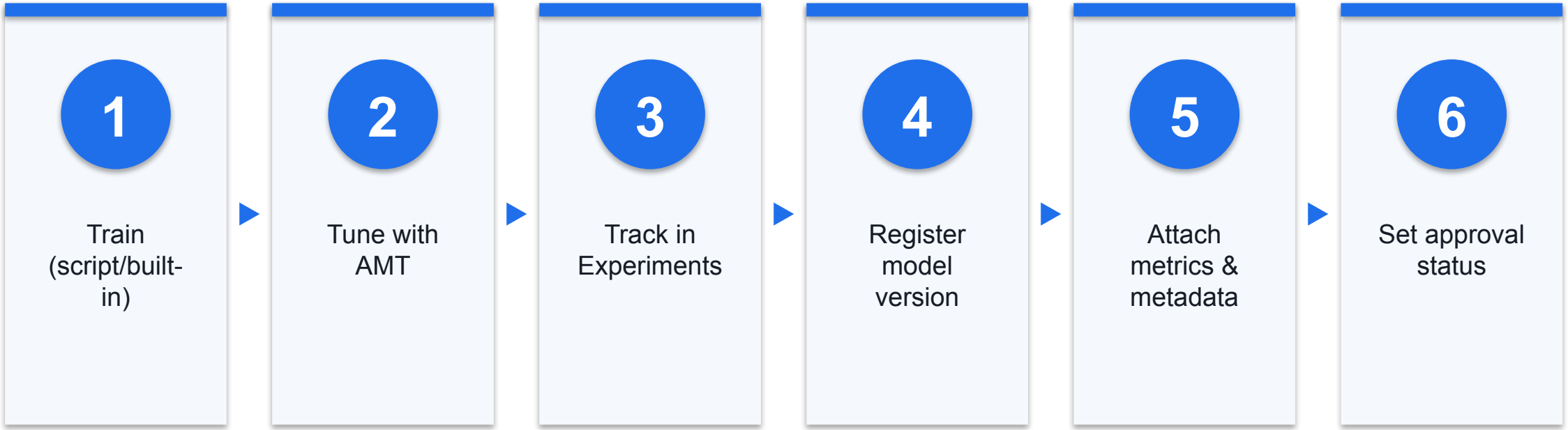
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Hyperparameter Tuning Strategies

Strategy	How it searches	Trade-off
Grid search	Every combination in a grid	Thorough but very expensive
Random search	Random points in the ranges	Cheap baseline, good coverage
Bayesian	Learns from prior trials	Sample-efficient; the usual default
Hyperband	Early-stops weak trials	Fast; great for many configurations

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Train → Register → Approve Workflow



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Evaluation Metrics — When to Use Which

Metric	Reads	Prefer when
Accuracy	Overall correct rate	Classes are balanced
Precision	Correctness of positive calls	False positives are costly (e.g. spam)
Recall (TPR)	Coverage of actual positives	False negatives are costly (e.g. disease)
F1	Harmonic mean of P and R	Imbalanced data; balance P vs R
ROC AUC	Ranking across thresholds	Comparing classifiers threshold-free
MAE / RMSE / R^2	Regression error / fit	Predicting a continuous value

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Diagnosing Model Fit

1. **Underfitting (high bias)**
   Poor on train and validation — add features/capacity.

2. **Overfitting (high variance)**
   Great on train, poor on validation — regularise/more data.

3. **Bias–variance trade-off**
   Balance the two for best generalisation.

4. **K-fold cross-validation**
   Rotate validation folds for a stable estimate.

5. **Regularisation**
   L1/L2, dropout, early stopping, pruning.

6. **Data leakage**
   Target/future info in features inflates scores — remove it.

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Bias & Explainability with Clarify

1. **Pre-training bias**
   Class imbalance and label skew before you train.
2. **Post-training bias**
   Fairness of predictions across groups.
3. **SHAP values**
   Per-prediction feature contributions.
4. **Global feature importance**
   Which features drive the model overall.
5. **Model Cards**
   Document intended use, data and known limitations.
6. **Why it matters**
   Trust, compliance and debugging of the model.

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Reading the Confusion Matrix

1. **True positive (TP)**
   Predicted positive and it was positive.

2. **False positive (FP)**
   Predicted positive but it was negative (false alarm).

3. **False negative (FN)**
   Predicted negative but it was positive (a miss).

4. **True negative (TN)**
   Predicted negative and it was negative.

5. **Precision = TP/(TP+FP)**
   Trust in a positive prediction.

6. **Recall = TP/(TP+FN)**
   Coverage of the actual positives.

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SageMaker Studio & Developer Tools

1

SageMaker Studio

Web IDE for notebooks, training and deployment.

2

Notebooks

Managed Jupyter for exploration and prototyping.

3

Training jobs

Run scalable training off the notebook.

4

Processing jobs

Preprocess and evaluate at scale.

5

Pipelines view

Author and monitor ML pipelines visually.

6

Model Registry view

Browse versions and approval status.

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WHAT YOU'LL DO

Hands-On Labs — ML Model Development

Labs 5–6

- Modelling Approach, Algorithms and Managed AI Services
- SageMaker Training Jobs and Debugger

Labs 7–8

- Automatic Model Tuning, Experiments and Model Registry
- Model Evaluation, Metrics, Bias and Explainability

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DOMAIN 02 · HANDS-ON BRIEFING

Modelling Approach, Algorithms and Managed AI…

LAB 5

Decide how to solve an ML problem before training: match problem types to approaches and compare SageMaker built-in algorithms, script-mode custom frameworks, JumpStart, Amazon Bedrock foundation models and managed AI services, then build a model/service selection matrix.

You'll build

A model/service selection matrix for five business scenarios with cost and interpretability notes.

Tools: Amazon SageMaker AI, SageMaker JumpStart, Amazon Bedrock, Rekognition, Comprehend, Textract, Translate, Transcribe

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LAB 5 · ARCHITECTURE & WORKFLOW

Workflow — Modelling Approach, Algorithms and…

1. List the ML problem types
2. Map each to an algorithm/service
3. Compare built-in vs script vs BYOC
4. Review JumpStart & Bedrock
5. Score with a selection matrix

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LAB 5 · KEY DECISIONS

Trade-offs — Modelling Approach, Algorithms and…

Managed AI service

- No training or ops to run
- Standard vision/text/speech tasks
- Fastest to integrate

Custom SageMaker model

- Full control and higher accuracy
- Trained on your own data/features
- More build and operations effort

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LAB 5 · VERIFY & TROUBLESHOOT

Modelling Approach, Algorithms and Managed AI…

✅ Test it — acceptance criteria

You can choose an AWS modelling approach for a given business outcome and justify it against data, cost, interpretability and operational requirements.

Troubleshooting

Unsure which algorithm fits	Match the problem type first (class/reg/cluster/forecast), then data size.
AI service accuracy too low for the domain	Fine-tune (JumpStart/Bedrock) or train a custom model.
Options hard to compare	Add cost, interpretability and ops columns to the selection matrix.

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DOMAIN 02 · HANDS-ON BRIEFING

SageMaker Training Jobs and Debugger

LAB 6

Run the training step end to end: configure input channels, instance, container and execution role, understand the training knobs and regularisation, choose input modes and cost options, and use SageMaker Debugger and Profiler to catch training problems early.

You'll build

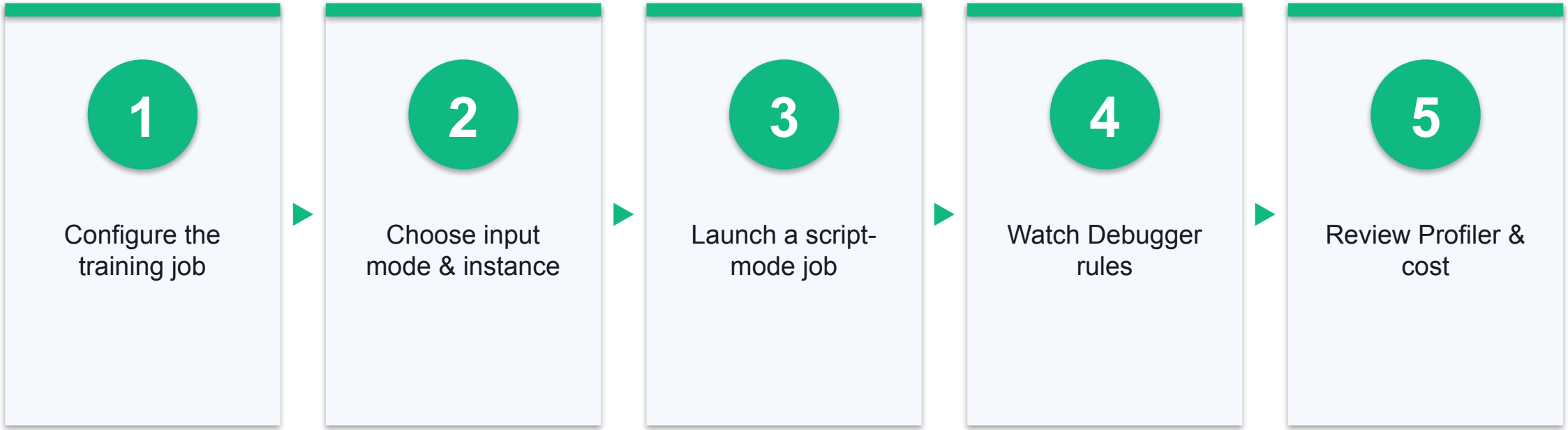
A configured, running training job plus Debugger/Profiler findings and a training-configuration checklist.

Tools: SageMaker Training, SageMaker Debugger, SageMaker Profiler, Amazon S3, Managed Spot training

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LAB 6 · ARCHITECTURE & WORKFLOW

Workflow — SageMaker Training Jobs and Debugger



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LAB 6 · KEY DECISIONS

Trade-offs — SageMaker Training Jobs and Debugger

Managed Spot training

- Up to ~90% cheaper
- Great for interruptible jobs
- Needs checkpointing to resume

On-demand training

- No interruption risk
- Simpler for short jobs
- Higher cost per hour

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LAB 6 · VERIFY & TROUBLESHOOT

SageMaker Training Jobs and Debugger

✅ Test it — acceptance criteria

You can configure and launch a SageMaker training job, apply regularisation and cost options, and use Debugger and the Profiler to diagnose training problems.

Troubleshooting

Training job fails to start	Check the execution role, input channels and instance limits.
Validation loss diverges	Lower the learning rate; add regularisation and early stopping.
Job is slow / GPU under-used	Use the Profiler to find I/O or utilisation bottlenecks.

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DOMAIN 02 · HANDS-ON BRIEFING

Automatic Model Tuning, Experiments and Model…

LAB 7

Make model development repeatable: design an Automatic Model Tuning plan and compare search strategies, track trials with SageMaker Experiments, and version models through the Model Registry with an approval workflow.

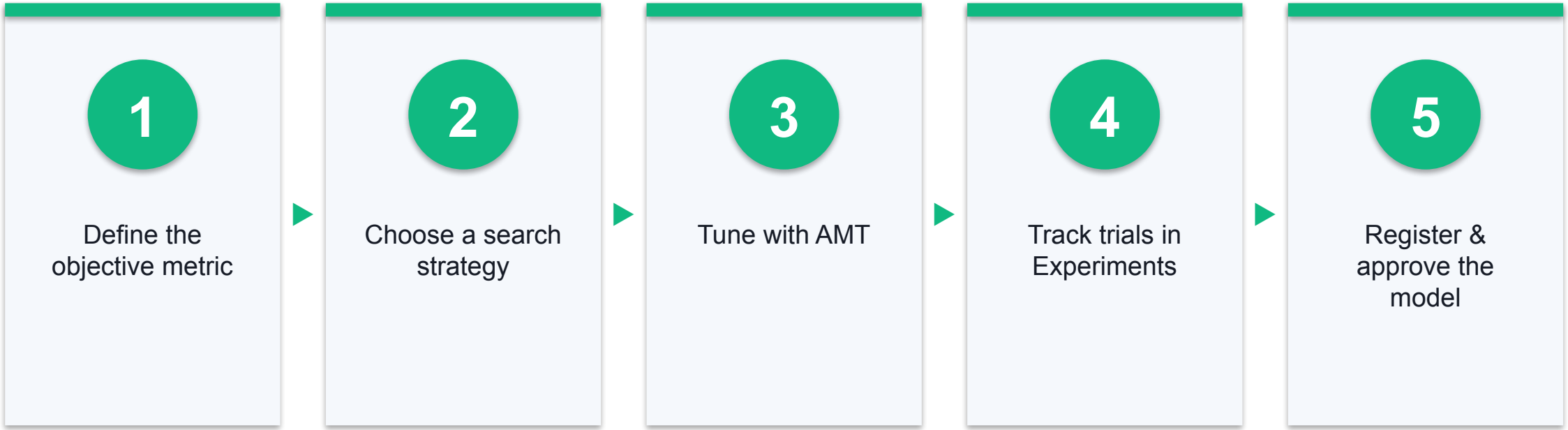
You'll build

A tuning plan, an experiment-tracking scheme and a model-versioning workflow from training job to approved model package.

Tools: SageMaker Automatic Model Tuning, SageMaker Experiments, SageMaker Model Registry, Amazon S3

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Workflow — Automatic Model Tuning, Experiments and…



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LAB 7 · KEY DECISIONS

Trade-offs — Automatic Model Tuning, Experiments…

Automatic Model Tuning

- Searches hyperparameters for you
- Bayesian/Hyperband are efficient
- Cost scales with job count

Manual tuning

- Full control of each run
- Cheaper for one or two knobs
- Slow and error-prone at scale

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LAB 7 · VERIFY & TROUBLESHOOT

Automatic Model Tuning, Experiments and Model…

✅ Test it — acceptance criteria

You can define a tuning plan, track trials in Experiments, and register and approve a model version in the Model Registry as a repeatable workflow.

Troubleshooting

Tuning is too expensive	Cap max/parallel jobs and narrow the hyperparameter ranges.
Best trial is hard to find	Track runs in SageMaker Experiments and sort by the objective metric.
Wrong model reaches deploy	Gate promotion with a Model Registry approval status.

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DOMAIN 02 · HANDS-ON BRIEFING

Model Evaluation, Metrics, Bias and Explainability

LAB 8

Decide whether a model is fit to promote: compare classification and regression metrics, diagnose overfitting/underfitting and leakage, review Clarify bias and SHAP explainability, and write an evaluation and acceptance checklist.

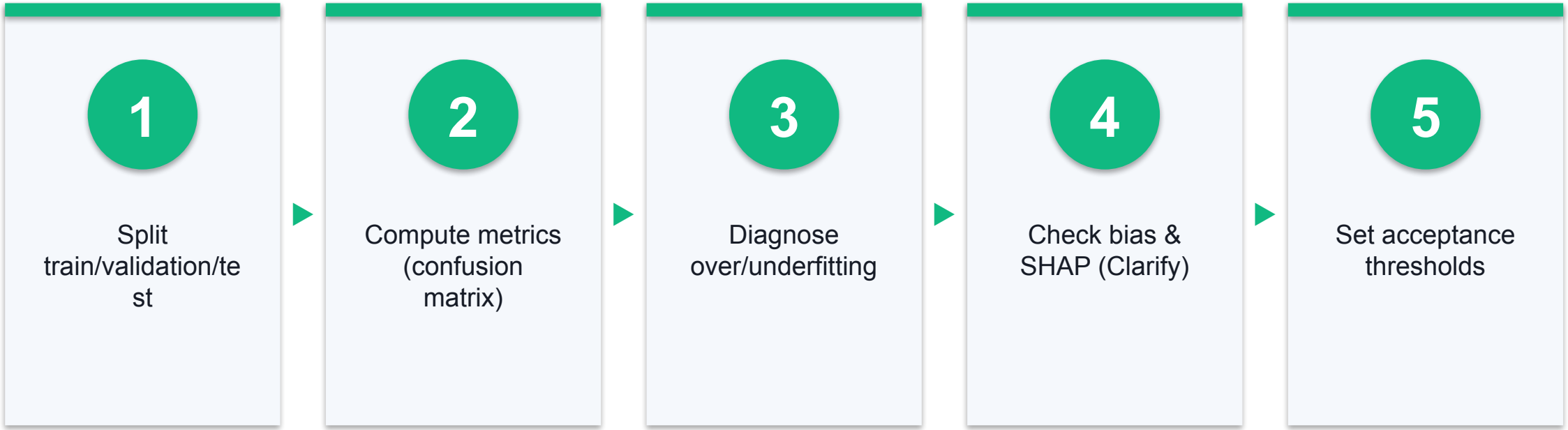
You'll build

An evaluation plan and model-acceptance checklist with metric thresholds and bias/explainability checks.

Tools: SageMaker Clarify, SageMaker training/evaluation, Amazon S3, Model Cards

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Workflow — Model Evaluation, Metrics, Bias and…



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LAB 8 · KEY DECISIONS

Trade-offs — Model Evaluation, Metrics, Bias and…

Optimise for precision

- Minimise false positives
- e.g. spam, fraud alerts
- May miss some true positives

Optimise for recall

- Minimise false negatives
- e.g. disease, safety screening
- May raise more false alarms

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LAB 8 · VERIFY & TROUBLESHOOT

Model Evaluation, Metrics, Bias and Explainability

✅ Test it — acceptance criteria

You can select appropriate metrics, interpret evaluation results, check bias and explainability, and decide whether a model is ready for approval or needs more data preparation or tuning.

Troubleshooting

High accuracy but poor real performance	Data is likely imbalanced or leaking — use F1 and recheck features.
Metrics differ sharply train vs test	Overfitting — regularise, add data or simplify the model.
Model unfair across groups	Use Clarify bias metrics and mitigate before approval.

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DOMAIN RECAP

Recap — ML Model Development

- You can now: choose an AWS modelling approach based on data, outcome, cost and operations (Domain 2: select a modelling approach)
- You can now: configure, run and debug a SageMaker training job efficiently (Domain 2: train models)
- You can now: run repeatable tuning, track experiments and version-control models (Domain 2: tune and manage models)
- You can now: choose metrics, interpret results and check bias and explainability before approval (Domain 2: evaluate model performance)

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03

DOMAIN 03

Deployment and Orchestration of ML Workflows

Inference infrastructure · endpoints & auto scaling · MLOps pipelines, CI/CD & IaC

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MLA-C01 EXAM WEIGHTING 22%

Key Concepts — Deployment and Orchestration of ML Workflows

1

Inference options

Real-time endpoints, serverless inference, asynchronous inference and batch transform — latency, payload, traffic and cost trade-offs.

2

Endpoint operations

Endpoint config, multi-model/multi-container endpoints, auto scaling on invocations-per-instance, blue/green and rollback.

3

Orchestration

SageMaker Pipelines steps (processing, training, tuning, condition, register, callback), Step Functions and EventBridge triggers.

4

CI/CD

CodePipeline and CodeBuild for automated build/test/deploy; model approval gates and artifact storage.

5

Infrastructure as Code

CloudFormation, CDK and SageMaker project templates to make ML infrastructure repeatable.

6

Safe delivery

Validation, approval, failure paths, retries and rollback so models move from development to production safely.

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SageMaker Inference Options

Option	Best for	Trade-off
Real-time endpoint	Low-latency online serving	Always-on cost per instance
Serverless inference	Spiky/intermittent traffic	Cold starts; scales to zero
Asynchronous inference	Large payloads, long inference	Queued, not instant
Batch transform	Offline scoring of a whole dataset	No live endpoint; run then stop

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Choosing an Inference Option

Low-latency online predictions 24×7	**Real-time endpoint + auto scaling**
Spiky, intermittent traffic	**Serverless inference**
Large payloads / long-running inference	**Asynchronous inference**
Score a whole dataset overnight	**Batch transform**
Many small models, cost-sensitive	**Multi-model endpoint**

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Endpoint Patterns & Safe Delivery

1. **Multi-model endpoint**
   Many models behind one endpoint — cost-efficient.

2. **Multi-container endpoint**
   Chain or host different containers/frameworks.

3. **Auto scaling**
   Scale on invocations-per-instance target metric.

4. **Blue/green & canary**
   Shift traffic gradually to a new model version.

5. **Shadow testing**
   Send live traffic copy to a candidate before promoting.

6. **Rollback path**
   Keep the previous version ready to restore instantly.

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Auto Scaling & Deployment Guardrails

1. **Target-tracking scaling**
   Scale on SageMakerVariantInvocationsPerInstance.
2. **Min/max capacity**
   Bound instance count for cost and availability.
3. **Cooldown periods**
   Avoid thrashing on brief traffic spikes.
4. **Health checks**
   Endpoint reports InService / Failed states.
5. **Deployment guardrails**
   Canary and linear traffic shifting with auto-rollback.
6. **CloudWatch alarms**
   Trigger rollback on latency/error thresholds.

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Containers & Infrastructure as Code

1

Docker + Amazon ECR

Package model/runtime; store the image in ECR.

2

Built-in vs BYOC

Use SageMaker containers or bring your own.

3

CloudFormation

Declarative templates for ML stacks.

4

AWS CDK

Define infrastructure in code (Python/TypeScript).

5

SageMaker Projects

MLOps templates that wire CI/CD end to end.

6

Version everything

Code, data and model artifacts under version control.

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Choosing an Orchestration Service

ML-native steps (process/train/tune/register)	**SageMaker Pipelines**
Coordinate AWS services with branching/retries	**AWS Step Functions**
Team already on Apache Airflow DAGs	**Amazon MWAA**
Build/test/deploy code and models (CI/CD)	**CodePipeline + CodeBuild**
Event-driven triggers (new data, model approved)	**Amazon EventBridge**

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SageMaker Pipelines vs Step Functions

Aspect	SageMaker Pipelines	Step Functions
Focus	ML-native lifecycle steps	Any AWS service workflow
Steps	Processing, train, tune, register	Lambda, ECS, SageMaker, choice
Lineage	Built-in ML lineage tracking	Manual
Best for	Standard train-to-deploy ML	Complex branching across services

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CI/CD for Machine Learning

1 **CodeCommit / Git**
Version model code and pipeline definitions.

2 **CodeBuild**
Build, unit-test and package containers.

3 **CodePipeline**
Chain source → build → deploy stages.

4 **Model approval gate**
Manual/automatic gate before production deploy.

5 **Artifact storage**
Store model.tar.gz and images in S3/ECR.

6 **EventBridge triggers**
Kick off retraining on new data or approval.

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MLOps Maturity

1

Level 0 — manual

Notebooks by hand; no automation.

2

Level 1 — pipeline

Automated training pipeline and Model Registry.

3

Level 2 — CI/CD

Automated build, test and deploy of models.

4

Monitoring loop

Drift detection triggers retraining.

5

Reproducibility

Versioned data, code, model and infra.

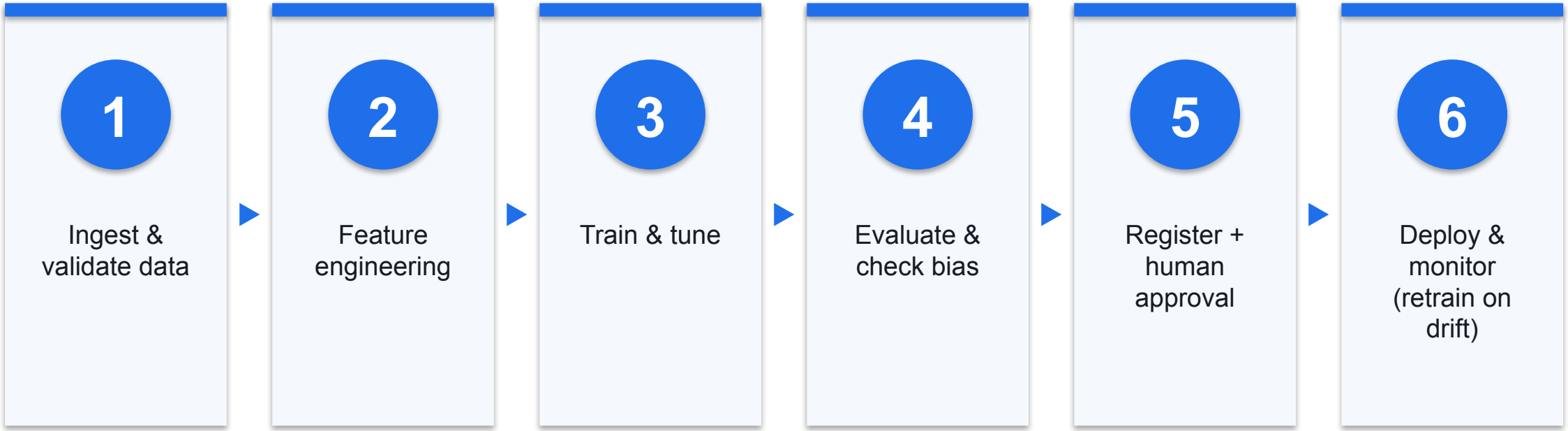
6

Governance

Approvals, lineage and audit throughout.

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A Gated MLOps Pipeline



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Real-time Endpoint Anatomy

1 **Model**
Artifact (model.tar.gz) plus inference container.

2 **Endpoint config**
Variants, instance type/count and weights.

3 **Production variants**
Split traffic for A/B or blue/green.

4 **Invoke API**
Clients call InvokeEndpoint over HTTPS.

5 **Auto scaling**
Add/remove instances on the target metric.

6 **Data capture**
Log requests/responses to S3 for monitoring.

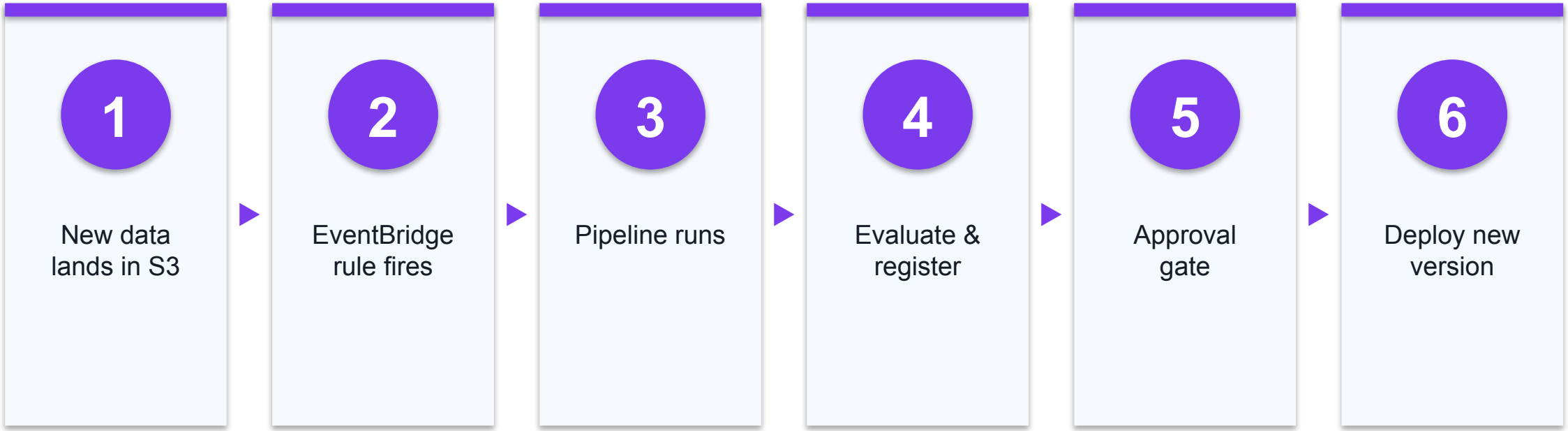
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Deployment Strategies Compared

Strategy	How it works	Best for
All-at-once	Replace the variant immediately	Dev/test, low risk
Canary	Shift a small % first, then the rest	Catch problems early in prod
Linear	Shift traffic in equal increments	Gradual, observable rollout
Blue/green	Stand up new, switch, keep old	Instant rollback capability
Shadow	Copy live traffic to a candidate	Validate before taking traffic

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Event-Driven Retraining



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WHAT YOU'LL DO

Hands-On Labs — Deployment and Orchestration of ML Workflows

Labs 9–9

- Deployment Options, Endpoints, Batch Transform and Auto Scaling

Labs 10–10

- MLOps Pipelines, CI/CD, Orchestration and IaC

—

- —

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DOMAIN 03 · HANDS-ON BRIEFING

Deployment Options, Endpoints, Batch Transform and…

LAB 9

Choose the right inference pattern: compare real-time endpoints, serverless inference, asynchronous inference and batch transform, understand endpoint configuration, auto scaling and safe delivery, and select infrastructure for several workloads.

You'll build

An inference-option comparison table with use-case deployment choices and an auto-scaling and rollback checklist.

Tools: SageMaker Inference, real-time & serverless endpoints, Batch Transform, Application Auto Scaling, CloudWatch

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Workflow — Deployment Options, Endpoints, Batch…

1. Pick the inference pattern
2. Create the endpoint config
3. Deploy the model
4. Register auto scaling
5. Verify metrics, then clean up

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LAB 9 · KEY DECISIONS

Trade-offs — Deployment Options, Endpoints, Batch…

Real-time endpoint

- Lowest latency, always available
- Good for online serving
- Pays for idle capacity

Serverless / batch

- Scales to zero or runs offline
- No idle cost
- Cold start / not instant

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LAB 9 · VERIFY & TROUBLESHOOT

Deployment Options, Endpoints, Batch Transform and…

✅ Test it — acceptance criteria

You can select a SageMaker deployment option from latency, traffic, payload and cost requirements, and configure auto scaling and a rollback path for an endpoint.

Troubleshooting

Endpoint stuck 'Creating' or fails	Check the container/model artifact and the CloudWatch logs.
Latency spikes under load	Enable auto scaling on the invocations-per-instance target.
Forgot to remove the endpoint	Delete the endpoint and config to stop hourly charges.

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DOMAIN 03 · HANDS-ON BRIEFING

MLOps Pipelines, CI/CD, Orchestration and IaC

LAB 10

Automate the ML lifecycle: compare SageMaker Pipelines, Step Functions, CodePipeline and EventBridge, design an end-to-end pipeline with approval and rollback gates, and identify where Infrastructure as Code and version control fit.

You'll build

An end-to-end MLOps pipeline design with approval, failure paths, rollback, CI/CD and IaC touchpoints.

Tools: SageMaker Pipelines, AWS Step Functions, CodePipeline, CodeBuild, EventBridge, CloudFormation/CDK

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Workflow — MLOps Pipelines, CI/CD, Orchestration…

1. Define the pipeline steps
2. Add a condition & register step
3. Add a human approval gate
4. Wire CI/CD and event triggers
5. Provision with IaC

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LAB 10 · KEY DECISIONS

Trade-offs — MLOps Pipelines, CI/CD, Orchestration…

SageMaker Pipelines

- ML-native processing/train/register steps
- Tight SageMaker integration
- Less generic than Step Functions

Step Functions

- Orchestrates any AWS service
- Rich branching, retries and waits
- More wiring to build

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LAB 10 · VERIFY & TROUBLESHOOT

MLOps Pipelines, CI/CD, Orchestration and IaC

✅ Test it — acceptance criteria

You can describe an automated ML workflow that moves models from development to production safely, with CI/CD, approval, retries, rollback and Infrastructure as Code.

Troubleshooting

A pipeline step fails midway	Add retries and inspect that step's CloudWatch logs.
A bad model reaches production	Add a condition step and a human approval gate before deploy.
Environment drifts between runs	Provision with CloudFormation/CDK and pin versions.

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DOMAIN RECAP

Recap — Deployment and Orchestration of ML Workflows

- You can now: select and operate SageMaker inference infrastructure for a workload (Domain 3: deploy models)
- You can now: design an automated, gated ML workflow from data to monitored production (Domain 3: orchestrate ML workflows)

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04

DOMAIN 04

ML Solution Monitoring, Maintenance and Security

Model & data monitoring · security & compliance · cost optimisation · exam readiness

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MLA-C01 EXAM WEIGHTING 24%

Key Concepts — ML Solution Monitoring, Maintenance and Security

1

Model monitoring

SageMaker Model Monitor: data capture, baselines, data quality, model quality, bias drift and feature-attribution drift.

2

Observability

CloudWatch metrics, logs, alarms, dashboards and anomaly detection; CloudTrail audit events for ML workloads.

3

Security

IAM least privilege and execution roles, KMS encryption, VPC endpoints, private subnets and network isolation.

4

Data protection

Data classification, masking, anonymisation, PII/PHI handling and data residency for ML datasets.

5

Cost optimisation

Right-sizing instances, endpoint auto scaling, serverless inference, batch transform, spot training and stopping idle notebooks.

6

Exam readiness

Mapping each lab to the four MLA-C01 domains and a timed practice set to identify weak topics before the exam.

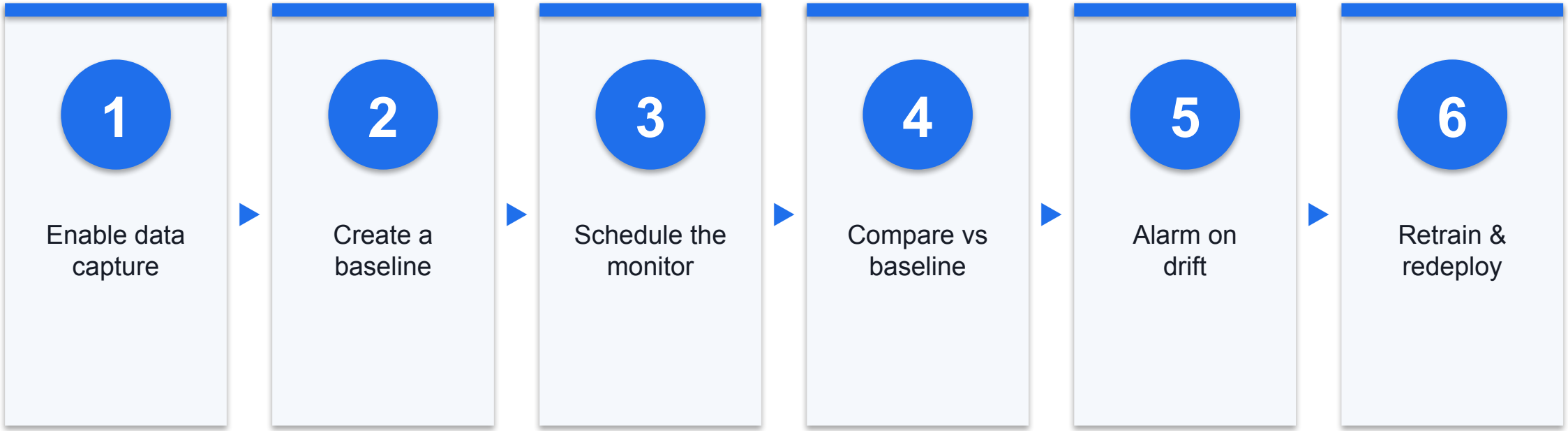
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Model Monitor — Four Drift Types

1. **Data quality drift**
   Live inputs diverge from the training baseline.
2. **Model quality drift**
   Accuracy/quality degrades vs ground truth.
3. **Bias drift**
   Fairness metrics move outside acceptable limits.
4. **Feature attribution drift**
   Feature importance shifts (Clarify + SHAP).
5. **Data capture**
   Endpoint logs inputs/outputs to S3 for analysis.
6. **Baselines & schedules**
   Compare captured data against a baseline on a schedule.

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Setting Up Model Monitoring



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Observability Stack for ML

1. **CloudWatch metrics**
   Invocations, latency, errors, instance utilisation.

2. **CloudWatch Logs & Insights**
   Endpoint/training logs; query with Logs Insights.

3. **CloudWatch alarms**
   Alert on latency, errors or drift thresholds.

4. **CloudTrail**
   Audit trail of every ML API call.

5. **AWS X-Ray**
   Trace requests across a serving pipeline.

6. **AWS Config**
   Track resource configuration and compliance.

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DOMAIN 04 · DEFENCE IN DEPTH

Security Layers for ML

1

IAM least privilege

Scoped policies and SageMaker execution roles.

2

KMS encryption

Encrypt S3, volumes and model artifacts with CMKs.

3

VPC endpoints / PrivateLink

Keep traffic off the public internet.

4

Private subnets & isolation

Run training/endpoints without internet egress.

5

Secrets Manager

Store credentials/tokens — never in notebooks.

6

Amazon Macie

Discover and classify PII in S3 datasets.

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Data Protection & Privacy

1. **Classification**
   Tag data by sensitivity (public → restricted).

2. **Masking / redaction**
   Hide PII fields before training or sharing.

3. **Anonymisation**
   Remove or hash identifiers; aggregate where possible.

4. **PII/PHI handling**
   Detect with Macie/Comprehend; encrypt with KMS.

5. **Data residency**
   Keep data in the required Region and account.

6. **Access logging**
   CloudTrail + S3 access logs for evidence.

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ML Governance

1. **Model Registry approval**
   Versioned packages with approval status.
2. **Model Cards**
   Document intended use, data and risk per model.
3. **ML Lineage Tracking**
   Trace data→training→model→endpoint.
4. **Model Dashboard**
   Single view of models, monitors and alerts.
5. **Role Manager**
   Build scoped ML personas quickly.
6. **Audit Manager**
   Evidence for compliance frameworks.

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Cost-Optimisation Levers

1. **Right-size instances**
   Match instance family/size to the workload.
2. **Auto scaling & serverless**
   Pay for demand, not idle capacity.
3. **Batch transform**
   Avoid always-on endpoints for offline scoring.
4. **Managed Spot training**
   Up to ~90% off with checkpointing.
5. **Multi-model endpoints**
   Consolidate many models onto shared instances.
6. **Budgets, tags & Compute Optimizer**
   Alert, allocate and right-size continuously.

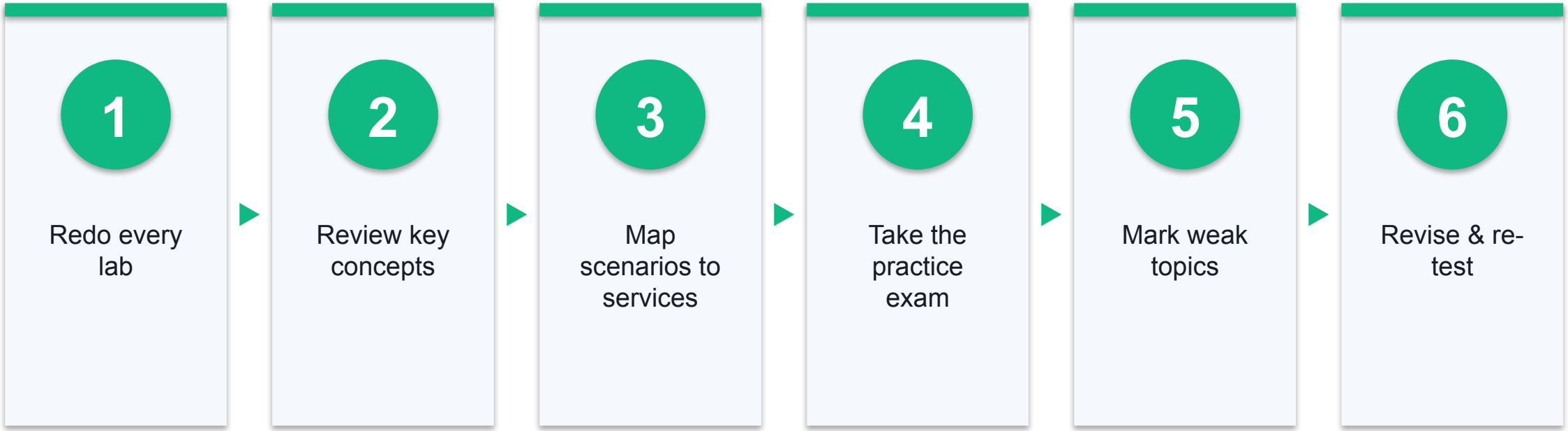
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Cost & Right-Sizing Decision Patterns

Interruptible training jobs	**Managed Spot training + checkpointing**
Endpoint idle most of the day	**Serverless inference or scale to zero**
Score millions of rows nightly	**Batch transform, no live endpoint**
Notebook left running overnight	**Auto-stop idle Studio apps**
Many small models per tenant	**Multi-model endpoint**

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Exam Readiness Loop



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CloudWatch Alarms & Dashboards for ML

1. **Endpoint metrics**
   Invocations, ModelLatency, OverheadLatency, errors.

2. **Instance metrics**
   CPU/GPU/memory/disk utilisation per instance.

3. **Alarms**
   Threshold and anomaly-detection alarms on key metrics.

4. **Dashboards**
   One pane for latency, errors, drift and cost.

5. **Logs Insights**
   Query endpoint/training logs for root cause.

6. **SNS actions**
   Notify or trigger automation when an alarm fires.

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Compliance & Audit for ML

1. **CloudTrail**
   Immutable record of every ML API call.

2. **AWS Config**
   Track configuration changes and compliance rules.

3. **Audit Manager**
   Collect evidence for frameworks continuously.

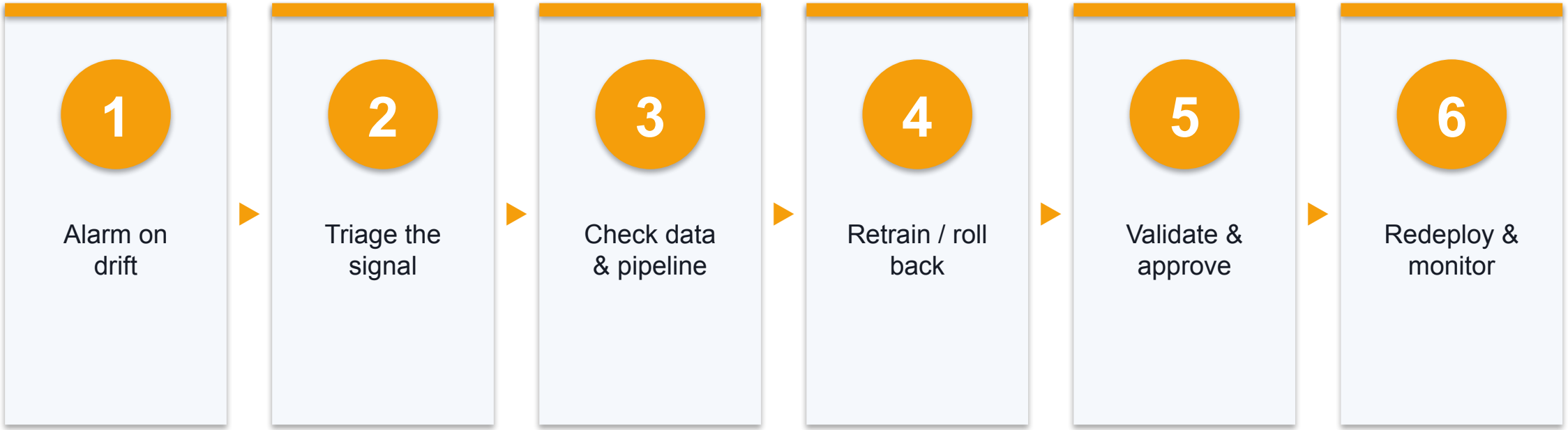
4. **Model Cards**
   Documented intended use, data and limitations.

5. **ML lineage**
   Trace data→training→model→endpoint.

6. **Tagging**
   Attribute cost and ownership across ML resources.

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Responding to a Drifting Model



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WHAT YOU'LL DO

Hands-On Labs — ML Solution Monitoring, Maintenance and Security

Labs 11–11

- Model Monitoring, Drift and Observability

Labs 12–12

- Security, Governance, Cost Optimisation and Exam Review

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DOMAIN 04 · HANDS-ON BRIEFING

Model Monitoring, Drift and Observability

LAB 11

Keep a production model healthy: enable endpoint data capture, baseline and schedule SageMaker Model Monitor for the four drift types, wire CloudWatch metrics, logs and alarms, and design the drift-to-retrain loop.

You'll build

A monitoring design with data capture, baselines, drift schedules, CloudWatch alarms and a drift-to-retrain plan.

Tools: SageMaker Model Monitor, SageMaker Clarify, Amazon CloudWatch, AWS CloudTrail, Amazon S3

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LAB 11 · ARCHITECTURE & WORKFLOW

Workflow — Model Monitoring, Drift and…

1. Enable endpoint data capture
2. Baseline the data
3. Schedule Model Monitor
4. Wire CloudWatch alarms
5. Plan drift-triggered retraining

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Trade-offs — Model Monitoring, Drift and…

Always-on monitoring

- Fast drift detection
- Continuous assurance
- Ongoing monitoring cost

Scheduled/periodic checks

- Lower cost
- Fine for stable models
- Slower to detect drift

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LAB 11 · VERIFY & TROUBLESHOOT

Model Monitoring, Drift and Observability

✅ Test it — acceptance criteria

You can enable data capture, baseline and schedule Model Monitor for the four drift types, wire CloudWatch alarms, and describe a drift-triggered retraining loop.

Troubleshooting

No monitoring data appears	Enable data capture on the endpoint before scheduling the monitor.
Monitor reports constant violations	Rebuild the baseline from representative recent data.
Alarms never fire	Check the CloudWatch metric, threshold and alarm state configuration.

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DOMAIN 04 · HANDS-ON BRIEFING

Security, Governance, Cost Optimisation and Exam…

LAB 12

Finish production-ready and exam-ready: apply IAM, KMS, VPC and data-protection controls, add governance with the Model Registry, Model Cards and lineage, build a cost-optimisation checklist, then map every lab to the four MLA-C01 domains and complete a timed practice set.

You'll build

Security, governance and cost-optimisation checklists, a lab-to-domain map and a timed practice-exam score with a weak-topic plan.

Tools: IAM, AWS KMS, Amazon VPC, Amazon Macie, SageMaker Model Registry/Model Cards, AWS Budgets, Cost Explorer

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Workflow — Security, Governance, Cost Optimisation…

1. Apply IAM/KMS/VPC controls
2. Protect PII/PHI data
3. Add governance & lineage
4. Build the cost checklist
5. Map labs to domains & practise

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Trade-offs — Security, Governance, Cost…

Tighter security & governance

- Least privilege, isolation, audit
- Lower breach and compliance risk
- More setup and review effort

Faster, looser access

- Quicker to build
- Fine only in a throwaway sandbox
- Unacceptable for real ML data

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LAB 12 · VERIFY & TROUBLESHOOT

Security, Governance, Cost Optimisation and Exam…

✅ Test it — acceptance criteria

You can secure, govern and cost-optimise an AWS ML workload, map each of the twelve labs to an exam domain, and produce a scored practice set with a weak-topic revision plan.

Troubleshooting

Access is over-permissive	Tighten IAM to least privilege and use scoped execution roles.
PII found in a training set	Detect with Macie/Comprehend; mask, anonymise and encrypt with KMS.
Surprise ML bill	Set Budgets and alarms; delete idle endpoints and notebooks.

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DOMAIN RECAP

Recap — ML Solution Monitoring, Maintenance and Security

- You can now: monitor a deployed model for drift and observe ML workloads in production (Domain 4: monitor and maintain ML solutions)
- You can now: secure, govern and cost-optimise ML in production and map every lab to the exam domains (Domain 4: secure, govern and optimise ML solutions)

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EXAM DECISION PATTERNS

Scenario → AWS Service Decision Maps

Reason from the requirement to the right AWS choice — the core skill the MLA-C01 exam tests.

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EXAM REASONING · DOMAIN 01

Data & Storage Decision Patterns

Store raw, prepared and feature data cheaply, securely	**S3 data lake with lifecycle + SSE-KMS**
Query terabytes in S3 with SQL, no servers	**Amazon Athena**
Near-real-time event ingestion into the lake	**Kinesis Data Firehose → S3**
Reusable low-latency features at serving time	**SageMaker Feature Store (online)**
Detect PII before training on a dataset	**Amazon Macie + SageMaker Clarify**

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EXAM REASONING · DOMAIN 01

Data Preparation & Labelling Patterns

No-code visual cleaning and profiling	**AWS Glue DataBrew**
Interactive ML feature engineering	**SageMaker Data Wrangler**
Label images/text at scale with review	**Ground Truth + Augmented AI (A2I)**
Automated rule checks on a dataset	**AWS Glue Data Quality**
Catalog schema for Athena and SageMaker	**AWS Glue crawler + Data Catalog**

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EXAM REASONING · DOMAIN 02

Modelling Decision Patterns

Tabular classification with strong accuracy	**XGBoost built-in algorithm**
No-code AutoML to find the best model fast	**SageMaker Autopilot**
Overfitting on training data	**Regularise, add data, early stopping**
Imbalanced fraud dataset — which metric?	**Precision/Recall/F1, not accuracy**
Explain why a model predicted an outcome	**SageMaker Clarify (SHAP)**

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EXAM REASONING · DOMAIN 03

Deployment Decision Patterns

Low-latency online predictions 24×7	**Real-time endpoint + auto scaling**
Spiky, intermittent traffic	**Serverless inference**
Score a large dataset overnight	**Batch transform**
Release a new model with zero downtime	**Blue/green or canary deployment**
Automate train→approve→deploy	**SageMaker Pipelines + approval gate**

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EXAM REASONING · DOMAIN 04

Monitoring, Security & Cost Patterns

Detect that live data drifted from training	**SageMaker Model Monitor**
Audit who called which ML API	**AWS CloudTrail**
Keep endpoint traffic off the internet	**VPC interface endpoints (PrivateLink)**
Cut training cost on interruptible jobs	**Managed Spot training**
Alert before the ML bill overruns	**AWS Budgets + cost allocation tags**

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EXAM REASONING · DOMAIN 03

Orchestration & MLOps Patterns

Standard ML train-to-deploy automation	**SageMaker Pipelines**
Branching workflow across many services	**AWS Step Functions**
Retrain automatically when new data lands	**EventBridge rule → pipeline**
Reproducible ML infrastructure	**CloudFormation / CDK / SageMaker Projects**
Gate a model before production	**Model Registry approval step**

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

Data Preparation

Built a secure S3 data lake and a raw-to-training preparation workflow with quality and bias checks.

2

Model Development

Selected approaches; trained, tuned, evaluated and registered models with SageMaker AI.

3

Deploy & Orchestrate

Chose inference infrastructure and designed a gated MLOps pipeline.

4

Monitor & Secure

Set up monitoring, security and cost optimisation for production ML.

5

Exam readiness

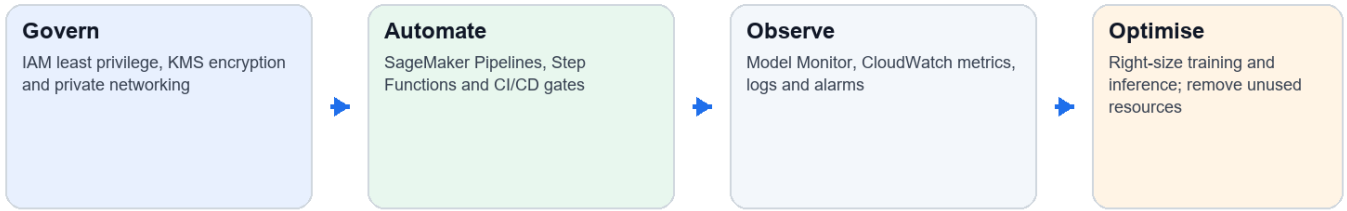
Mapped every lab to an MLA-C01 domain and completed a timed practice set.

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OPERATE SAFELY

Governance & Monitoring in Production

Management and Governance Loop



All diagrams are aligned to the twelve AWS hands-on labs and reused across the PPT, Learner Guide and Lesson Plan.

Govern, secure and audit ML workloads continuously in production.

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KEEP ML HEALTHY & EFFICIENT

Monitoring & Cost Optimisation Loop

Cost and Monitoring Lifecycle



All diagrams are aligned to the twelve AWS hands-on labs and reused across the PPT, Learner Guide and Lesson Plan.

Monitor drift and spend, then right-size and retrain — a continuous loop.

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NEXT STEPS

Preparing for the MLA-C01 Exam

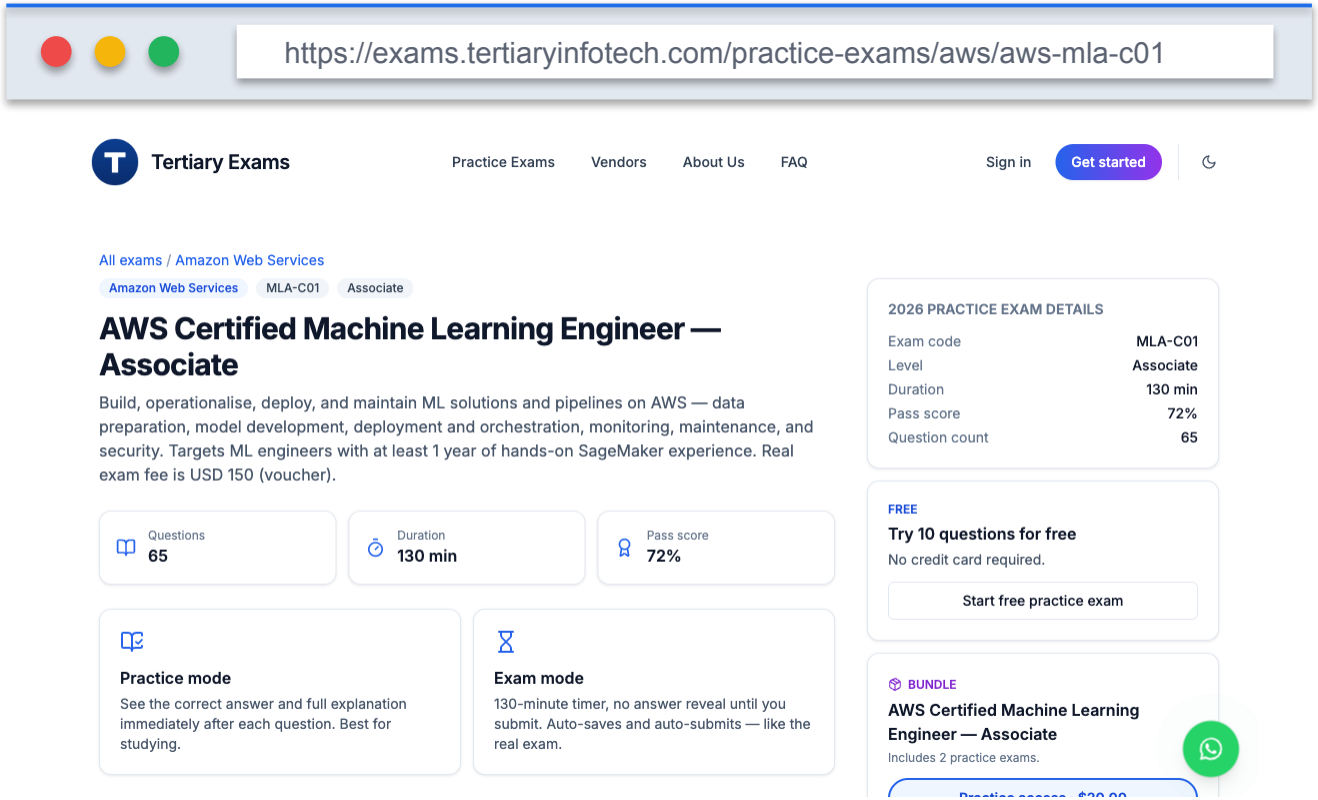
- Redo every lab until the AWS console and CLI workflow is automatic.
- Review the Key Concepts and the 'Test it' check for each lab and domain.
- Practise mapping each scenario to the right AWS service and the four exam domains.
- Take the AWS official MLA-C01 practice question set and the Tertiary practice exam.
- Book the exam via Pearson VUE or PSI (online proctoring or a test centre).

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TEST YOURSELF · EXAMS PORTAL

Practice Exam

1. Open the Tertiary Infotech MLA-C01 practice exam.
2. Attempt it under timed exam conditions.
3. Review the explanation for every question.
4. Revisit any lab whose domain you miss.
5. Re-take until you consistently pass.



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WRAP-UP

Assessment

- Written Assessment (SAQ) — 1 hour. Case Study (CS) — 1 hour.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at https://lms-tms.tertiaryinfotech.com/.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then Case Study (CS) — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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Digital Attendance (Mandatory)

1

Mandatory each session

Take AM, PM and Assessment digital attendance for WSQ-funded courses.

2

TRAQOM QR code

The trainer displays the SSG digital-attendance QR code (TRAQOM).

3

Scan to submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

At least 75% attendance is required to be eligible for assessment and funding.

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GOOD LUCK WITH MLA-C01

Thank You!

You are now ready to build ML solutions on AWS — and to sit the AWS Certified Machine Learning Engineer - Associate (MLA-C01) exam.

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